

HW 1

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1 Problem Formulation

Consider the linear system ($Ax = b$). Instead of solving it directly, we formulate it as an optimization problem by minimizing the norm:

$$L(x) = \min_x \|Ax - b\| \quad (1)$$

2 Tasks

2.1 1) Gradient Descent Algorithm

We aim to solve the optimization problem:

$$\min_x \|Ax - b\|^2 \quad (2)$$

where:

- A is an $m \times n$ matrix.
- x is an $n \times 1$ vector (our unknown variable).
- b is an $m \times 1$ vector.

2.1.1 Gradient Computation

The loss function is:

$$L(x) = \|Ax - b\|^2 = (Ax - b)^T(Ax - b) \quad (3)$$

Taking its gradient with respect to x :

$$\nabla_x L(x) = 2A^T(Ax - b) \quad (4)$$

2.1.2 Gradient Descent Update Rule

Gradient Descent (GD) updates x iteratively as:

$$x^{(k+1)} = x^{(k)} - \alpha \cdot \nabla_x L(x^{(k)}) \quad (5)$$

where:

- α is the learning rate.
- $\nabla_x L(x) = 2A^T(Ax - b)$ is the gradient.

To ensure convergence, a good choice for α is:

$$0 < \alpha < \frac{1}{\lambda_{\max}(A^T A)} \quad (6)$$

where $\lambda_{\max}(A^T A)$ is the largest eigenvalue of $(A^T A)$.

2.2 2) Alternative Methods for Rank-Deficient A

If A is not full-rank, solving $Ax = b$ directly may lead to instability. Here are five alternative methods:

1. **Gaussian Elimination with Partial Pivoting** Uses row exchanges for numerical stability. Eliminates coefficients to form an upper triangular matrix, then solves using back substitution.
2. **Singular Value Decomposition (SVD)** Factorizes A as:

$$A = U\Sigma V^T \quad (7)$$

where Σ contains singular values. The least-squares solution is:

$$x = V\Sigma^{-1}U^T b \quad (8)$$

3. **QR Factorization** Decomposes A as:

$$A = QR \quad (9)$$

where Q is orthogonal and R is upper triangular. The system reduces to:

$$Rx = Q^T b \quad (10)$$

which is solved via back substitution.

4. **LU Decomposition with Pivoting** Factorizes A as:

$$PA = LU \quad (11)$$

where P is a permutation matrix. Solves two triangular systems sequentially.

5. **Tikhonov Regularization (Ridge Regression)** Stabilizes the solution by adding a penalty term:

$$\min_x \{ \|Ax - b\|^2 + \lambda \|x\|^2 \} \quad (12)$$

The solution is:

$$x = (A^T A + \lambda I)^{-1} A^T b \quad (13)$$

3 References

For all algorithm searches, refer to Wikipedia. To check the text, use ChatGPT, except for **Gaussian Elimination with Partial Pivoting**, for which you should use this source. For **Regularization Methods – Tikhonov Regularization**, refer to this document.

***For more information check the report.md file**